Trigonometry

ANSWER KEY

Section 1: Exact Trig Ratios & Reference Angles

1 Find the exact value of $\sin(210^\circ)$.

a. $-\frac{1}{2}$ ✓

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. $\frac{\sqrt{3}}{2}$

210° is in Quadrant III with reference angle 30°. Sine is negative in QIII, so $\sin(210^\circ) = -\sin(30^\circ) = -1/2$.

2 Find the exact value of $\cos\left(\frac{3\pi}{4}\right)$.

a. $-\frac{\sqrt{2}}{2}$ ✓

b. $\frac{\sqrt{2}}{2}$

c. $-\frac{1}{2}$

d. $\frac{\sqrt{3}}{2}$

$3\pi/4$ is in QII where cosine is negative: $\cos(3\pi/4) = -\cos(\pi/4) = -\sqrt{2}/2$.

3 If θ is reflex and $\tan(\theta) = \frac{4}{3}$, find the exact value of $\sin(\theta)$.

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $\frac{3}{5}$

Since $\tan(\theta) > 0$ and θ is reflex, θ lies in Quadrant III. Here $x = -3, y = -4, r = 5$. Thus $\sin(\theta) = -4/5$.

Section 2: Fundamental Trigonometric Identities

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4 Simplify the expression $(1-\cos^2\theta)(1+\cot^2\theta)$ for all valid θ.

a. **1** ✓

b. $\sin^2\theta$

c. $\tan^2\theta$

d. $\sec^2\theta$

$1-\cos^2\theta=\sin^2\theta$. $1+\cot^2\theta=\csc^2\theta=1/\sin^2\theta$. Product is $\sin^2\theta\times(1/\sin^2\theta)=1$.

5 Prove the identity $\dfrac{\sin\theta}{1-\cos\theta}+\dfrac{1-\cos\theta}{\sin\theta}\equiv 2\csc\theta$. What is the common numerator after combining fractions?

a. $2(1-\cos\theta)$ ✓

b. $\sin^2\theta+\cos^2\theta$

c. $2\sin\theta$

d. $1-2\cos\theta$

$$\frac{\sin^2\theta+(1-\cos\theta)^2}{\sin\theta(1-\cos\theta)}=\frac{\sin^2\theta+1-2\cos\theta+\cos^2\theta}{\sin\theta(1-\cos\theta)}=\frac{2-2\cos\theta}{\sin\theta(1-\cos\theta)}=\frac{2(1-\cos\theta)}{\sin\theta(1-\cos\theta)}=2\csc\theta.$$

6 Given $\cos\theta=-\frac{5}{13}$ and $90^\circ<\theta<180^\circ$, find $\tan\theta$.

a. $-\frac{12}{5}$ ✓

b. $\frac{12}{5}$

c. $-\frac{5}{12}$

d. $\frac{13}{12}$

In QII, sine is positive: $\sin\theta=\sqrt{1-25/169}=12/13$. Then $\tan\theta=\frac{12/13}{-5/13}=-12/5$.

Section 3: Solving Trigonometric Equations

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7 Solve $2\sin(2x) = 1$ for $0^\circ \le x \le 180^\circ$.

a. $x = 15^\circ$ **and** $x = 75^\circ$ ✓

b. $x = 30^\circ$ and $x = 150^\circ$

c. $x = 15^\circ$ and $x = 105^\circ$

d. $x = 45^\circ$ and $x = 135^\circ$

$\sin(2x) = 1/2$. For $0^\circ \le 2x \le 360^\circ$: $2x = 30^\circ$ or $150^\circ \Rightarrow x = 15^\circ$ or 75°.

8 Solve $2\cos^2\theta - 5\cos\theta + 2 = 0$ for $0^\circ \le \theta \le 360^\circ$.

a. $\theta = 60^\circ$ **and** $\theta = 300^\circ$ ✓

b. $\theta = 30^\circ$ and $\theta = 330^\circ$

c. $\theta = 60^\circ$ and $\theta = 120^\circ$

d. $\theta = 45^\circ$ and $\theta = 315^\circ$

Factor: $(2\cos\theta - 1)(\cos\theta - 2) = 0$. Since $\cos\theta \le 1$, $\cos\theta = 1/2 \Rightarrow \theta = 60^\circ$ or $360^\circ - 60^\circ = 300^\circ$.

9 Solve $3\tan(x) = 2\sin(x)$ for $0 < x < 2\pi$.

a. $x = \pi$ **(excluded) or** $\cos x = 3/2$ **(no solution)** ✓

b. No solutions in $(0, 2\pi)$

c. $x = \pi/3$ and $5\pi/3$

d. $x = \pi/6$ and $5\pi/6$

$3\frac{\sin x}{\cos x} - 2\sin x = 0 \Rightarrow \sin x(3 - 2\cos x) = 0$. $\sin x = 0$ gives $x = \pi$ (since domain is open interval $(0, 2\pi)$, $x = \pi$ is a root!). $2\cos x = 3 \Rightarrow \cos x = 1.5$ impossible. Root is $x = \pi$.

Section 4: Trigonometric Graphs & Transformations

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10 State the amplitude and period of the function $y = 3\sin(2x) - 1$.

a. **Amplitude** $= 3$**, Period** $= \pi$ ✓

b. Amplitude $= 3$, Period $= 2\pi$

c. Amplitude $= 2$, Period $= \pi$

d. Amplitude $= 4$, Period $= \pi$

Amplitude is $|a| = 3$. Period is $2\pi/b = 2\pi/2 = \pi$.

11 Find the range of the function $f(x) = 4 - 3\cos(x)$ for all real x.

a. $[1, 7]$ ✓

b. $[-1, 7]$

c. $[1, 4]$

d. $[-3, 3]$

Since $-1 \le \cos(x) \le 1$: minimum is $4 - 3(1) = 1$, maximum is $4 - 3(-1) = 7$. Range is $[1, 7]$.

12 Find the number of solutions to the equation $\sin(2x) = \cos(x)$ in the interval $0 \le x \le 2\pi$.

a. **4 solutions** ✓

b. 2 solutions

c. 3 solutions

d. 6 solutions

$2\sin x\cos x - \cos x = 0 \Rightarrow \cos x(2\sin x - 1) = 0$. $\cos x = 0 \Rightarrow x = \pi/2, 3\pi/2$.
$\sin x = 1/2 \Rightarrow x = \pi/6, 5\pi/6$. Total 4 solutions.

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